



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING IIT (BHU)

Computer Systems Organization (Code: CSO 211)

Time: 24 hours

End Term Examination

Maximum Marks: 40

All answers should be written in a clear and concise manner. Plagiarism will be checked, and if detected, marks will be deducted. The maximum marks related to the questions are indicated at the end of the questions. Make suitable assumptions for incomplete information.

Q1. Explain in brief the following terms related to computer architecture. [$5 \times 1 = 5$]

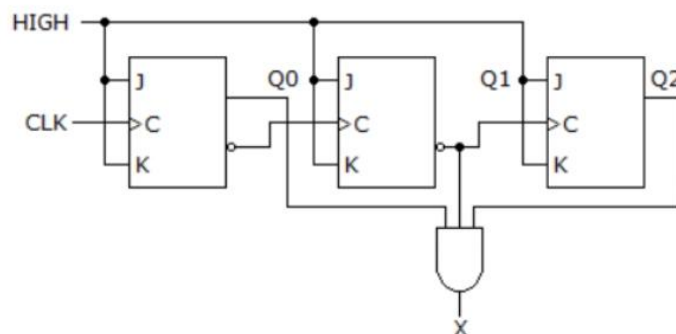
- i) Addressing modes
- ii) Registers
- iii) Buses
- iv) Memory
- v) Interrupt

Q2. An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contains the number 200. Evaluate the effective address if the addressing mode of the instruction is (i) Direct (ii) Immediate (iii) Relative (iv) Register indirect (v) Index with R1 as the index register [$5 \times 1 = 5$]

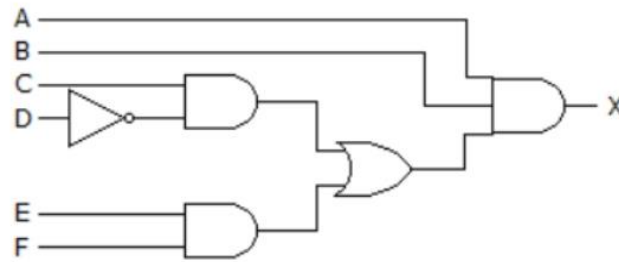
Q3. A computer uses a memory unit 256k words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts : indirect bit , an operation code , a register code part to specify one of 64 registers, and an address part. [$2+1+2=5$]

- i) How many bits are there in the operation code , the register code part , and the address part?
- ii) Draw the instruction word format and indicate the number of bits in each part.
- iii) How many bits are there in the data and address inputs of the memory?

Q4. Construct the truth table for the below given circuit considering Q0 Q1 Q2 as input and X as output. [5]



Q5. Convert the output of the below given circuit in sum-of-products (SOP) form [5]



Q6. Minimize the following function in SOP minimal form using K-Maps: $F(A, B, C, D) = m(1, 2, 6, 7, 8, 13, 14, 15) + d(3, 5, 12)$ where d denotes the don't care condition. [5]

Q7. Consider the following program segment for a hypothetical CPU having three user registers R1, R2 and R3. Let the memory be byte addressable with size 32 bits, and the program has been loaded starting from memory location 1000 (decimal). If an interrupt occurs while the CPU has been halted after executing the HALT instruction, the return address (in decimal) saved in the stack will be [5]

Instruction	Operation	Instruction size (in words)
MOV $R_1, 5000$	$R_1 \leftarrow \text{Memory}[5000]$	2
MOV $R_2, (R_1)$	$R_2 \leftarrow \text{Memory}[(R_1)]$	1
ADD R_2, R_3	$R_2 \leftarrow R_2 + R_3$	1
MOV 6000, R_2	$\text{Memory}[6000] \leftarrow R_2$	2
HALT	Machine Halts	1

Q8. Find the expression for X in the given circuit [5]

